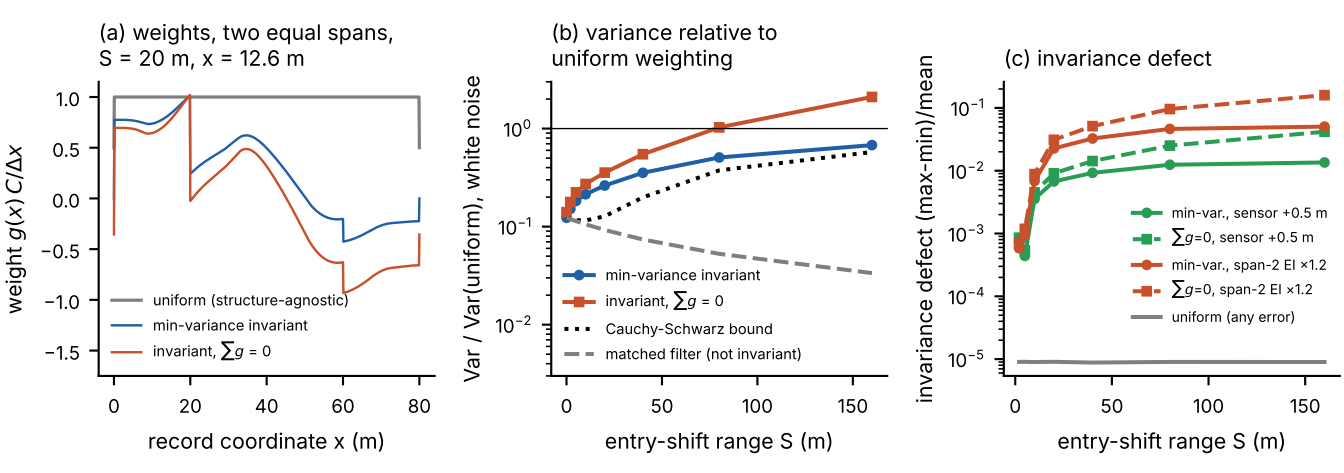
(a) weights, two equal spans,
S = 20 m, x = 12.6 m
weight $g(x)\,C/\Delta x$
record coordinate x (m)
uniform (structure-agnostic)
min-variance invariant
invariant, $\sum g$ = 0
(b) variance relative to
uniform weighting
Var / Var(uniform), white noise
entry-shift range S (m)
min-variance invariant
invariant, $\sum g$ = 0
Cauchy-Schwarz bound
matched filter (not invariant)
(c) invariance defect
invariance defect (max-min)/mean
entry-shift range S (m)
min-var., sensor +0.5 m
$\sum g$=0, sensor +0.5 m
min-var., span-2 EI ×1.2
$\sum g$=0, span-2 EI ×1.2
uniform (any error)